



MEDIACREST
TRAINING COLLEGE

Mediacrest Training College

Nairobi, Kenya

Advanced Data Analytics & Machine Learning

8-Week Advanced Training Program

Comprehensive Session Notes

Week 2 · Day 2

Regression Analysis, Diagnostic Testing & Economic Elasticities

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“A model is only as good as its diagnostic checks. Never trust an OLS output without validating its underlying assumptions.”

About This Document

These comprehensive notes accompany **Week 2, Day 2** of the **Advanced Data Analytics & Machine Learning** program at Mediacrest Training College. Following our foundational work in data preprocessing, outlier detection, and feature engineering, we now transition into the core of inferential statistics and predictive modeling: **Regression Analysis and Diagnostic Testing**.

This session bridges classical statistical theory with applied econometrics and machine learning. We will explore how to quantify relationships between variables, build predictive models, and, most importantly, rigorously test the assumptions that make those models valid. The datasets used range from classic automotive benchmarks (`mtcars`) to clinical hypertension records, macroeconomic time series, and microeconomic demand functions.

Course at a Glance

	8 Weeks (20+ intensive sessions)
Duration	
	Python (pandas, NumPy, SciPy, statsmodels, scikit-learn), R
Core Tools	
	Inferential statistics, OLS regression, diagnostic testing, econometric modeling
Focus	
	Production-ready ML system with diagnostics, explainability & policy recommendations
Capstone	
	Healthcare, Finance, Macroeconomics (CBK, KNBS), and Microeconomic Demand
Context	

Where This Session Fits

Week 2 is titled “**Advanced Feature Engineering, Dimensionality Reduction & Model Alignment**.”

- Day 1:** Outlier Detection, Feature Encoding Strategies, Scaling, and Algorithm Sensitivity.
- Day 2 (this document):** Simple & Multiple Linear Regression, Diagnostic Testing (VIF, Cook’s D, Normality, Homoscedasticity, Independence), and Elasticity Modeling.
- Day 3:** Feature Selection, Dimensionality Reduction (PCA), and building predictive baselines.

How to Use These Notes

- Follow along in code.** Every listing is runnable. Open the companion Jupyter Notebook and execute each cell as you read.
- Boxes matter.** **Key Concept** boxes hold the theory; **Warning** boxes highlight common pitfalls; **Kenyan Context** boxes connect material to local industry realities; **Mathematical Foundation** boxes provide rigorous statistical proofs.

- **Do the exercises.** Each chapter ends with practice tasks designed to test your statistical intuition.

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1 Foundations of Regression and Correlation

1.1 What is Regression Analysis?

Regression analysis is a powerful statistical technique used to examine the relationship between one or more **independent variables** (predictors, features, or X) and a **dependent variable** (outcome, target, or Y). It helps us understand how changes in the independent variables are associated with changes in the dependent variable.

Regression serves three primary purposes in data science and econometrics:

1. **Prediction:** Forecasting continuous outcomes (e.g., predicting house prices, fuel efficiency, or future sales).
2. **Inference:** Identifying which factors significantly influence the outcome and quantifying the magnitude of their effect.
3. **Control:** Understanding the marginal effect of a single variable while holding all other confounding variables constant (*ceteris paribus*).

1.2 History of Regression Analysis

The term “regression” has a fascinating historical origin rooted in biology, not economics or computer science.

- **Sir Francis Galton (Late 1800s):** Coined the term while studying the heights of parents and their children. He observed that extreme heights in parents tended to “regress toward the mean” in their offspring. Tall parents had children who were taller than average, but shorter than the parents themselves.
- **Karl Pearson & Ronald Fisher (Early 20th Century):** Formalized the mathematical foundations of correlation and regression, integrating them into the broader framework of inferential statistics, hypothesis testing, and experimental design.
- **Modern Era:** Regression has evolved from a purely statistical tool to a foundational algorithm in machine learning (e.g., Linear Regression, Ridge/Lasso), econometrics, and biostatistics.

1.3 Correlation vs. Causation

The Golden Rule of Statistics

Correlation does not imply causation.

- **Correlation:** A statistical measure describing the degree to which two variables move together. It quantifies *association* (strength and direction), ranging from -1 to $+1$.
- **Causation:** Implies that a change in one variable *directly produces* a change in another. Establishing causation requires rigorous experimental design (e.g., Randomized Controlled Trials) or advanced causal inference techniques.
- **The Trap:** Two variables may be highly correlated due to:

1. **Direct causation:** A causes B .
2. **Reverse causation:** B causes A .
3. **Confounding:** A hidden third variable C causes both A and B (e.g., ice cream sales and shark attacks are both driven by summer weather).
4. **Spurious correlation:** Pure coincidence with no logical connection.

2 Exploratory Data Analysis & Correlation

Before fitting any regression model, we must explore the data visually and statistically. We will use the classic `mtcars` dataset, extracted from the 1974 *Motor Trend US* magazine, comprising fuel consumption and 10 aspects of automobile design and performance for 32 automobiles.

2.1 Loading and Inspecting the Data

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 import scipy.stats as stats
6 import statsmodels.api as sm
7 import statsmodels.formula.api as smf
8 from statsmodels.stats.outliers_influence import variance_inflation_factor
9 from statsmodels.stats.diagnostic import het_breuschpagan
10 from statsmodels.stats.stattools import durbin_watson
11 import warnings
12
13 warnings.filterwarnings('ignore')
14 sns.set_theme(style="whitegrid")
15 pd.set_option('display.float_format', '{:.4f}'.format)
16
17 # Load data
18 url_mtcars = "https://raw.githubusercontent.com/vincentarelbundock/Rdatasets/
19             master/csv/datasets/mtcars.csv"
19 mtcars = pd.read_csv(url_mtcars, index_col=0)
20 print(mtcars.head())
```

Listing 1: Loading libraries and the mtcars dataset

2.2 Visualizing Relationships: Scatter Plots

Scatter plots are **essential** before running any regression or correlation analysis. They reveal the shape of the relationship (linear, curvilinear, or none) and highlight potential outliers.

```
1 plt.figure(figsize=(8, 6))
2 sns.regplot(x='wt', y='mpg', data=mtcars,
3             scatter_kws={'color': 'mcbblue', 's': 30},
4             line_kws={'color': 'mcaccent'})
5 plt.title("Relationship between Weight and MPG")
6 plt.xlabel("Weight (1000 lbs)")
7 plt.ylabel("Miles per Gallon")
8 plt.show()
```

Listing 2: Scatter plot with regression line

Observation: There is a strong negative linear correlation. As the weight of the car increases, the miles per gallon decreases.

2.3 Pearson vs. Spearman Correlation

Hypothesis Testing for Correlation

Null Hypothesis (H_0): There is no significant correlation between the variables ($\rho = 0$).

Alternative Hypothesis (H_1): There is a significant correlation ($\rho \neq 0$).

Decision Rule: If the p-value < 0.05, we reject the null hypothesis.

```
1 # Pearson Correlation (Linear relationship, assumes normality)
2 r_pearson, p_pearson = stats.pearsonr(mtcars['wt'], mtcars['mpg'])
3 print(f"Pearson r: {r_pearson:.5f}, p-value: {p_pearson:.2e}")
4
5 # Spearman Correlation (Monotonic relationship, rank-based, robust to outliers)
6 r_spearman, p_spearman = stats.spearmanr(mtcars['wt'], mtcars['mpg'])
7 print(f"Spearman rho: {r_spearman:.5f}, p-value: {p_spearman:.2e}")
```

Listing 3: Computing Pearson and Spearman correlations

Output:

Pearson r: -0.86766, p-value: 1.29e-10

Spearman rho: -0.88642, p-value: 1.49e-11

Interpretation

Both tests show a **very strong, highly significant negative correlation**. The Spearman coefficient is slightly stronger, suggesting a highly consistent monotonic decrease in MPG as weight increases, robust to any extreme outliers. We decisively reject H_0 .

2.4 Correlation and Covariance Matrices

To understand the broader feature space, we compute pairwise matrices for multiple continuous variables simultaneously.

```
1 num_vars = mtcars[['mpg', 'cyl', 'disp', 'hp', 'wt']]
2 corr_matrix = num_vars.corr(method='pearson')
3 print(corr_matrix.round(2))
```

Listing 4: Correlation Matrix

Insight: `wt` and `disp` are highly correlated (0.89). Including both in a multiple regression model may cause **multicollinearity**, which we must diagnose later.

3 Simple Linear Regression (SLR)

3.1 The SLR Concept and Equation

The Simple Linear Regression Model

The population model is defined as:

$$Y = \beta_0 + \beta_1 X + \epsilon \quad (1)$$

Where:

- Y : Dependent variable (Outcome, e.g., mpg).
- X : Independent variable (Predictor, e.g., wt).
- β_0 : Y-intercept (Expected value of Y when $X = 0$).
- β_1 : Slope coefficient (Expected change in Y for a 1-unit increase in X).
- ϵ : Error term (Residuals), assumed to be normally distributed with mean 0 and constant variance σ^2 .

3.2 Fitting the SLR Model

```
1 model_slr = smf.ols('mpg ~ wt', data=mtcars).fit()  
2 print(model_slr.summary())
```

Listing 5: Fitting SLR: MPG predicted by Weight

Key Output Interpretation:

- **Intercept** ($\beta_0 = 37.285$): If a car weighed 0 lbs (a physical impossibility), its MPG would be 37.28. The intercept is often just a mathematical anchor.
- **Slope** ($\beta_1 = -5.344$): For every 1,000 lbs increase in weight, MPG **decreases** by 5.344 units.
- **P>|t|** (< 0.001): The weight variable is highly statistically significant.
- **R-squared** (0.753): 75.3% of the variance in MPG is explained solely by the vehicle's weight.
- **F-statistic** (91.38, $p < 0.001$): The overall model is significantly better than a naive model that just predicts the mean MPG.

3.3 Confidence Intervals vs. Prediction Intervals

When forecasting a new observation (e.g., a car weighing 3,500 lbs), we must distinguish between two types of intervals:

```
1 new_car = pd.DataFrame({'wt': [3.5]})  
2 pred_slr = model_slr.get_prediction(new_car).summary_frame(alpha=0.05)
```

```
$ print(pred_slr[['mean', 'mean_ci_lower', 'mean_ci_upper', 'obs_ci_lower', 'obs_ci_upper']])
```

Listing 6: Predicting for a new car (wt = 3.5)

Interval Definitions

- **Confidence Interval (Mean):** Estimates where the *average* MPG of *all* 3,500 lb cars will fall. It is narrower because it only accounts for the uncertainty in estimating the mean.
- **Prediction Interval (Observation):** Estimates where a *single, specific* 3,500 lb car's MPG will fall. It is wider because it accounts for both the uncertainty in the mean *and* the random variance (ϵ) of individual cars.

3.4 Real-World Application: Hypertension Data

In clinical data, missing values are common. We load `htn_dat.csv`, perform complete case analysis (dropping NAs), and model Systolic Blood Pressure (SBP) using Body Mass Index (BMI).

```
1 # Assuming 'htn' is loaded and cleaned via htn.dropna()
2 model_htn = smf.ols('SBP ~ BMI', data=comp_data).fit()
3 print(model_htn.summary())
```

Listing 7: SLR on Hypertension Data

Result: $SBP = 92.84 + 0.857(BMI)$. For every 1 unit increase in BMI, SBP increases by 0.857 mmHg ($p < 0.001$). However, the R-squared is only 0.041, indicating that while the relationship is statistically significant, BMI alone explains very little of the total variance in blood pressure (which is highly multifactorial).

4 Multiple Linear Regression (MLR)

4.1 The MLR Concept

Simple linear regression is rarely sufficient in real-world scenarios because outcomes are influenced by multiple factors. Multiple Linear Regression extends SLR by including k predictors.

The Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \epsilon \quad (2)$$

The core interpretation shifts to the **ceteris paribus** condition: β_i represents the expected change in Y for a 1-unit increase in X_i , *holding all other variables constant*.

4.2 Fitting the MLR Model

Let's predict mpg using both wt (weight) and hp (horsepower).

```
1 model_mlr = smf.ols('mpg ~ wt + hp', data=mtcars).fit()
2 print(model_mlr.summary())
```

Listing 8: Fitting MLR: MPG predicted by Weight and Horsepower

Key Output Interpretation:

- **Weight** ($\beta_1 = -3.878$): A 1,000 lb increase decreases MPG by 3.88 units, *holding horsepower constant*. (Notice this is less extreme than the SLR coefficient of -5.344 , because SLR was absorbing the effect of horsepower).
- **Horsepower** ($\beta_2 = -0.032$): A 1 unit increase in HP decreases MPG by 0.032 units, *holding weight constant*.
- **Adjusted R-squared** (0.815): By adding hp, the explained variance increased from 75.3% to 81.5%. Adjusted R-squared is preferred in MLR because it penalizes the addition of irrelevant predictors, preventing overfitting.

5 Regression Diagnostics: The LINE Assumptions

Ordinary Least Squares (OLS) regression relies on strict assumptions. If violated, coefficients may be biased, standard errors incorrect, and p-values invalid. We use the acronym **LINE** to remember them.

5.1 1. Linearity

The relationship between X and Y must be linear. This is primarily checked via scatter plots of Y vs. X and residuals vs. fitted values.

5.2 2. Independence of Errors

Residuals must be independent of each other (no autocorrelation). This is crucial in time-series data.

- **Test:** Durbin-Watson Statistic.
- **Rule of Thumb:** $DW \approx 2$ indicates no autocorrelation. $DW < 1.5$ suggests positive autocorrelation; $DW > 2.5$ suggests negative autocorrelation.
- **Our Result:** $DW = 1.36$. Slight positive autocorrelation, but often acceptable in small cross-sectional datasets.

5.3 3. Normality of Residuals

The error terms (ϵ) should be normally distributed. This is critical for valid hypothesis testing (p-values and confidence intervals), especially in small samples.

- **Tests:** Shapiro-Wilk and Anderson-Darling.
- **Our Result:** Shapiro-Wilk $W = 0.928$, $p = 0.034$. We technically reject the null hypothesis of normality. However, visual inspection of the Q-Q plot shows only minor deviations at the tails, which is often tolerable in moderate samples.

```
1 residuals = model_mlr.resid
2 stat_sw, p_sw = stats.shapiro(residuals)
3 print(f"Shapiro-Wilk Test: W={stat_sw:.5f}, p-value={p_sw:.5f}")
```

Listing 9: Normality Testing

5.4 4. Equal Variance (Homoscedasticity)

The variance of residual errors must be constant across all levels of the predictors. If the variance changes (e.g., a “funnel” shape in residual plots), it is called **heteroscedasticity**.

- **Test:** Breusch-Pagan Test.
- **Hypotheses:** H_0 : Residuals have constant variance. H_1 : Heteroscedasticity is present.
- **Our Result:** $BP = 0.88$, $p = 0.64$. We fail to reject H_0 . Homoscedasticity is satisfied.

Consequence of Heteroscedasticity

If heteroscedasticity is present, the OLS coefficients remain unbiased, but their **standard errors are incorrect**, rendering t-tests and p-values unreliable. The professional fix is to use **Robust Standard Errors** (e.g., HC3).

5.5 5. No Multicollinearity

Independent variables should not be highly correlated with each other. Multicollinearity inflates standard errors, making coefficients unstable and statistically insignificant.

- **Test:** Variance Inflation Factor (VIF).
- **Rule of Thumb:** $VIF > 5$ (or 10) indicates severe multicollinearity. $VIF = 1$ indicates no correlation.

```
1 X_vif = mtcars[['wt', 'hp']].copy()
2 X_vif['const'] = 1 # VIF requires the constant term
3 vif_data = pd.DataFrame()
4 vif_data["feature"] = X_vif.columns
5 vif_data["VIF"] = [variance_inflation_factor(X_vif.values, i) for i in range(
    X_vif.shape[1])]
6 print(vif_data)
```

Listing 10: Calculating VIF

Result: VIF for both `wt` and `hp` is 1.77. This is well below the threshold of 5, indicating no problematic multicollinearity.

5.6 Handling Influential Observations (Cook's Distance)

Some data points may exert disproportionate leverage on the regression line due to extreme X values (leverage) and large residuals.

- **Test:** Cook's Distance.
- **Threshold:** Observations with Cook's Distance $> \frac{4}{N}$ are considered highly influential.
- **Action:** In our `mtcars` model, 4 cars were flagged: *Chrysler Imperial*, *Fiat 128*, *Toyota Corolla*, *Maserati Bora*.

Best Practice for Outliers in Regression

Do *not* blindly delete outliers. Investigate them first. If they are data entry errors, remove them. If they are genuine extreme cases, consider robust regression techniques (e.g., RANSAC, Huber Regressor) or report the model both with and without the influential points to show transparency.

6 Macroeconomic Modeling and Elasticities

In economics, we are often less interested in absolute unit changes and more interested in **percentage changes** (elasticities).

6.1 The Limitation of Linear Models in Economics

A standard linear model assumes that a 1-unit change in the Exchange Rate always leads to a fixed unit change in the Consumer Price Index (CPI), regardless of the current level of prices. This is economically unrealistic. Variables like CPI, Exchange Rates, and GDP grow exponentially over time.

6.2 The Log-Log (Double-Log) Model

By taking the natural logarithm of both the dependent and independent variables, we linearize exponential relationships and the coefficients directly represent **elasticities**.

The Elasticity Model

$$\ln(Y) = \beta_0 + \beta_1 \ln(X_1) + \beta_2 \ln(X_2) + \epsilon \quad (3)$$

In this specification, β_1 represents the **elasticity** of Y with respect to X_1 . **Definition:** A 1% increase in X_1 leads to a $\beta_1\%$ change in Y , holding other variables constant.

6.3 Modeling CPI: Exchange Rates and Interest Rates

Let's model the Kenyan Consumer Price Index (CPI) against the Exchange Rate (KES/USD) and Lending Interest Rates.

```
1 # Assuming df_macro contains 'CPI', 'ExchRate', and 'LendIntRates'
2 model_cpi_log = smf.ols('np.log(CPI) ~ np.log(ExchRate) + np.log(LendIntRates)',
3                           data=df_macro).fit()
3 print(model_cpi_log.summary())
```

Listing 11: Log-Log Model for CPI

Output Interpretation:

- **Equation:** $\ln(\text{CPI}) = -0.704 + 1.515 \ln(\text{ExchRate}) - 0.552 \ln(\text{LendIntRates})$
- **R-squared:** 0.935 (The log-log model explains 93.5% of the variance, a significant improvement over the linear model's 77.2%).
- **Exchange Rate Elasticity (1.515):** Holding lending rates constant, a 1% depreciation in the currency is associated with a 1.515% increase in CPI. Since $1.515 > 1$, CPI is **elastic** with respect to the exchange rate, indicating strong “pass-through” effects of imported inflation.
- **Interest Rate Elasticity (−0.552):** Holding exchange rates constant, a 1% increase in lending rates is associated with a 0.552% decrease in CPI. Since $|-0.552| < 1$, the relationship is **inelastic**. Higher rates reduce inflation, but the effect is dampened, requiring substantial rate hikes by the Central Bank of Kenya (CBK) to achieve modest reductions in CPI.

Macroeconomic Policy Implications

The high exchange rate elasticity (1.515) confirms that Kenya, as a net importer of fuel, machinery, and intermediate goods, is highly vulnerable to currency depreciation. Monetary policy (interest rate hikes) has an inelastic, lagged effect on inflation, suggesting that exchange rate stability is a more immediate prerequisite for price stability than aggressive rate hikes alone.

7 Microeconomic Demand Analysis

The double-log model is the standard specification for estimating demand functions, as it allows us to extract all key elasticities simultaneously.

7.1 The Demand Model Specification

Quantity demanded (Q) is a function of own price (P), consumer income (Y), the price of a substitute good (P_s), and the price of a complementary good (P_c).

```
1 model_demand = smf.ols('np.log(Quantity) ~ np.log(Price) + np.log(Income) + np.  
    log(SubPrice) + np.log(CompPrice)', data=df_demand).fit()  
2 print(model_demand.summary())
```

Listing 12: Double-Log Demand Model

Output Interpretation (Elasticities):

1. **Own-Price Elasticity** ($E_p = -1.139$): A 1% increase in price decreases quantity demanded by 1.139%. Since $|E_p| > 1$, demand is **elastic**. Consumers are highly responsive to price changes; a price increase would reduce total revenue.
2. **Income Elasticity** ($E_Y = 0.753$): A 1% increase in consumer income increases quantity demanded by 0.753%. Since $0 < E_Y < 1$, the product is a **normal necessity** (e.g., basic foodstuffs, utilities). Demand grows as consumers get wealthier, but at a slower rate than income.
3. **Cross-Price Elasticity - Substitute** ($E_{Sub} = 0.537$): A 1% increase in the price of a substitute good increases demand for this product by 0.537%. Because $E_{Sub} > 0$, the goods are **substitutes** (e.g., Safaricom and Airtel airtime).
4. **Cross-Price Elasticity - Complement** ($E_{Comp} = -0.407$): A 1% increase in the price of a complement decreases demand for this product by 0.407%. Because $E_{Comp} < 0$, the goods are **complements** (e.g., smartphones and mobile data bundles).

8 Keynesian Economics and the Consumption Function

8.1 The Theory

Coined by **John Maynard Keynes** in his seminal 1936 work, *The General Theory of Employment, Interest and Money*, the Absolute Income Hypothesis states that consumption depends primarily on current disposable income.

The Fundamental Psychological Law

"Men are disposed, as a rule and on average, to increase their consumption as their income increases, but not by as much as the increase in their income."

$$C = \alpha + \beta Y \quad (4)$$

Where:

- C : Aggregate Consumption
- Y : Aggregate Income
- α : Autonomous consumption (consumption when income is zero, funded by debt or savings)
- β : **Marginal Propensity to Consume (MPC)**

8.2 Estimating the MPC

The MPC is the proportion of an additional unit of income that is spent on consumption: $MPC = \frac{\Delta C}{\Delta Y}$. Theoretical bounds dictate that $0 < MPC < 1$.

```
1 # Standard SLR
2 model_cons = smf.ols('Consumption ~ Income', data=df_income).fit()
3 print(model_cons.summary())
4
5 # Long-Run Model (No Intercept / Forced through origin)
6 model_cons_lr = smf.ols('Consumption ~ Income - 1', data=df_income).fit()
7 print(model_cons_lr.summary())
```

Listing 13: Estimating the Consumption Function

Interpretation:

- **Standard Model:** Consumption = $11.37 + 0.898 \times \text{Income}$. The empirical **MPC = 0.898**. For every additional \$1 billion of income, consumers spend \$898 million and save \$102 million. This perfectly validates Keynes's law ($0 < 0.898 < 1$).
- **Long-Run Model:** In the long run, economic theory dictates that if aggregate income is zero, consumption must be zero. Forcing the intercept to zero yields Consumption = $0.905 \times \text{Income}$. Both models yield an $R^2 \approx 0.999$, but the choice depends on whether we are modeling short-run dynamics (with autonomous consumption) or long-run equilibrium.

The Multiplier Effect in Fiscal Policy

The Keynesian multiplier is defined as $k = \frac{1}{1-MPC}$. With an MPC of 0.898, $k = \frac{1}{1-0.898} \approx 9.8$. This implies that a KES 1 billion injection of government spending (e.g., infrastructure projects) will theoretically generate KES 9.8 billion in total GDP growth through successive rounds of spending. This highlights the potency of fiscal stimulus in consumer-driven economies.

9 Best Practices, Checklist & Deliverables

9.1 Best Practices for Regression Modeling

- Visualize First:** Always plot scatter plots and correlation heatmaps before fitting models. Anscombe’s Quartet proves that summary statistics can be deeply misleading.
- Check Assumptions (LINE):** Never trust regression output blindly. Test for Linearity, Independence, Normality, and Equal Variance.
- Choose the Right Functional Form:** Use Log-Log models for elasticities and Level-Level models for marginal effects.
- Beware of Multicollinearity:** Check VIF. If predictors are highly correlated, consider Principal Component Analysis (PCA) or dropping redundant variables.
- Context is King:** A statistically significant coefficient must make theoretical sense (e.g., ensuring MPC is between 0 and 1, or verifying the Law of Demand).

9.2 Professional Preprocessing & Diagnostics Checklist

Stage	Action	Rationale
EDA	Scatter plots & Correlation matrices.	Reveals non-linearities and multicollinearity early.
Model Selection	SLR for baseline, MLR for control.	MLR isolates unique effects (ceteris paribus).
Multicollinearity	Check $VIF < 5$.	Prevents inflated standard errors and unstable coefficients.
Influential Points	Cook’s Distance $> 4/n$.	Identifies points that disproportionately dictate the regression line.
Normality	Shapiro-Wilk & Q-Q Plots.	Validates the mathematical foundation of OLS inference.
Homoscedasticity	Breusch-Pagan Test.	Ensures standard errors are reliable; use Robust SE if violated.
Independence	Durbin-Watson Test.	Critical for time-series; $DW \approx 2$ is the target.
Elasticities	Use Log-Log transformations.	Directly yields percentage-change interpretations.

Table 1: The Professional Regression Diagnostics Checklist.

9.3 Session Deliverables

Final Deliverable: The Diagnostic Memo

Draft a 1-page technical memo addressed to the Chief Economist. Your memo must include:

1. **Model Specification:** Justify the use of the Log-Log model over the standard linear model for the CPI dataset.
2. **Diagnostic Validation:** Report the VIF, Breusch-Pagan, and Durbin-Watson statistics, explicitly stating whether the LINE assumptions are satisfied and what remedial actions (if any) were taken.
3. **Economic Interpretation:** Translate the coefficients of the CPI model into plain-English policy recommendations regarding exchange rate management and interest rate adjustments by the Central Bank.

10 Glossary of Key Terms

Ceteris Paribus	A Latin phrase meaning "all other things being equal." In MLR, it is the assumption that all other predictors are held constant when interpreting a single coefficient.
Elasticity	The percentage change in the dependent variable resulting from a 1% change in the independent variable. Directly obtained from Log-Log model coefficients.
Heteroscedasticity	The condition where the variance of the error terms is not constant across all levels of the independent variables, violating an OLS assumption.
Homoscedasticity	The desirable condition where the variance of the error terms is constant.
Marginal Propensity to Consume (MPC)	The proportion of an additional unit of income that is spent on consumption ($0 < MPC < 1$).
Multicollinearity	A situation where two or more independent variables in a regression model are highly correlated, inflating standard errors.
Ordinary Least Squares (OLS)	A method for estimating the unknown parameters in a linear regression model by minimizing the sum of the squares of the differences between observed and predicted values.
Variance Inflation Factor (VIF)	A measure of how much the variance of an estimated regression coefficient is increased because of multicollinearity.

11 References & Further Reading

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